

An Intelligent Ride-Sharing Recommendation Method Based on Graph Neural Network and Evolutionary Computation

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Abstract—This research is dedicated to addressing user recommendation matching and multi-objective optimization problems in ride-sharing services. For addressing the challenge of node classification in social networks, the Graph Attention Network with Opinion Dynamics (OD-GAT) is proposed. This model combines opinion dynamics and attention mechanism, which can make full use of multi-dimensional information for social relationship reasoning, and at the same time simulate the influence of individuals by other objects in the group, realize more accurate prediction and reasoning of social relationships, improved service quality and ride-sharing safety. To address the intricate task of balancing multiple objectives, including average detour cost, average response rate, and average user similarity rate, we introduce a novel evolutionary computation method for optimizing ride-sharing scenarios. This approach tackles the dynamic ride-sharing matching problem by emphasizing human factors in the optimization goals, successfully overcoming challenges related to local optima and convergence. Experimental validation confirms the effectiveness of OD-GAT in feature extraction and classification, showcasing the method's fastest convergence speed and global optimum achievement across three key metrics.

Index Terms—Graph attention network, opinion dynamics, social network analysis, evolutionary computation.

I. INTRODUCTION

WITH the increasing penetration rate of private cars, while offering convenience, it has led to rising travel costs, traffic congestion, and environmental pollution, issues that will worsen as car numbers grow [1]. Ride-sharing, which allows drivers to share unused vehicle space with passengers on similar routes, enhances road resource utilization, reducing travel costs and mitigating pollution [2]. However, safety

concerns in ride-sharing remain due to poor management and supervision, with incidents affecting public safety [3]. Implementing user categorization based on attributes like age and gender could improve safety, particularly for vulnerable groups such as solo women and minors.

Despite the advantages of ride-sharing in improving resource utilization and reducing environmental impact, passenger safety remains a challenge. User data in online social networks is often encrypted through various cloud computing schemes [4], [5], and location privacy protection in the Internet of Vehicles (IoV) ensures vehicles do not expose real-time location information [6], [7]. Extracting meaningful features from limited public data is impractical, and dynamic ride-sharing poses a multi-objective optimization problem, where high costs, poor real-time performance, and local optima are common issues [15]. Thus, analyzing passengers' social relationships and classifying them before matching is essential to enhance service quality and safety [8].

Online Social Networks (OSNs) provide abundant user node and relationship data, representing snapshots of network structures formed through social interactions. This data holds significant theoretical and practical value in Social Network Analysis (SNA), especially for public security, where analyzing users' social networks is a key method in intelligence gathering [9], [10]. The advent of Graph Neural Networks (GNNs) has further advanced network modeling, enhancing the prediction and reasoning of social relationships by leveraging graph-structured data, with attention mechanisms often introduced for more complex relational tasks [11]. In the context of ride-sharing, the matching problem can be viewed as a multi-objective optimization challenge [12], which seeks to balance conflicting goals. Evolutionary computation methods, such as genetic algorithms, simulate biological evolution through operations like crossover and mutation, iteratively evolving optima that balance multiple objectives, offering decision makers a range of optimized solutions [13], [14].

Existing researches have some shortcomings. First of all, most researches on dynamic ride-sharing matching algorithms only consider the efficiency factor [15], [16], [17], [18], and do not consider passengers' safety and experience through analyzing social network; secondly, a large number of researches are limited to the analysis and mining of a single social network, lacking mutual assistance, verification and fusion

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of multi-dimensional information [20], [22], [36]; and the existing attention mechanism is difficult to simulate and learn the spread of opinions and popularity in the community, and cannot estimate the magnitude of the mutual influence of individuals in the decision-making process [23], [25], [36]. Finally, in the multi-objective optimization of dynamic ride-sharing, only a very small number of researches have adopted the evolutionary computation method [40], which leads to problems such as local optima and difficult convergence during the process.

In view of the above problems, the contributions of our research are as follows:

- 1) The Graph Attention Network with Opinion Dynamics (OD-GAT) is proposed, which can make full use of multi-dimensional information for social relationship reasoning, and at the same time, simulate the influence of individuals by other objects in the group, realizing more accurate prediction and reasoning of relationships;
- 2) A novel evolutionary computation method for ride-sharing is proposed to unify the framework of dynamic ride-sharing problems. Compared with traditional dynamic ride-sharing optimization problems, the optimization objective of humanity factors of user similarity is considered in this research. The problem of falling into local optima and convergence difficulty in the process is addressed;
- 3) The experimental validation confirmed the feature extraction and classification performance of OD-GAT, as well as the effectiveness of utilizing evolutionary computation for multi-objective optimization. The experiments indicated that our method possesses the fastest convergence speed and achieves the global optimum across three key metrics.

II. RELATED WORK

A. Dynamic Ride-Sharing Matching

Dynamic ride-sharing matching is a multi-objective optimization problem, with varying objectives based on emphasis. Common optimization objectives include detour cost ξ_{Δ} , response rate R_{res} , and user similarity rate R_{sim} . User similarity is a humanity factor that involves the classification of social network nodes, requiring analysis and processing of online social network. Many studies have focused only on efficiency issues [15], [16], [17], [18] and have not addressed user similarity. Additionally, online social network often lacks attribute information, making it challenging to learn and process the complex network topology.

Early algorithms primarily focused on path planning and efficiency improvement, without considering humanity factors such as user similarity. As a result, user safety and ride experience were not improved. Shen et al. [15] proposed an optimized filtering and refining algorithm, which utilized heuristic methods and other optimization techniques in the context of big data. It also employed request grouping pre-processing, but did not consider user similarity for the sake of computational simplicity. Armant and Brown [16] introduced a mixed integer programming optimization algorithm aiming

TABLE I
COMPARISON OF CLASSICAL DYNAMIC RIDE-SHARING METHODS

Method	Detour Cost	Response Rate	User Similarity
F&R [15]	Y	Y	N
MDD [16]	Y	N	N
PA [18]	Y	Y	N
MA [39]	Y	Y	Y
SAGA [40]	Y	Y	N
BBO [42]	Y	Y	Y

to minimize detour costs within a time window. This method emphasized computational performance and was suitable for handling high-density ride requests, but it did not address the response rate and user similarity under random ride-hailing probabilities. Kleiner et al. [18] proposed an auction-based incentive-compatible dynamic ride-sharing algorithm, which aimed to minimize the total travel distance through fixed assignments while maximizing the response rate.

Few studies have incorporated humanity factors into the optimization objectives of the ride-sharing process to enhance safety and ride experience. Lyu et al. [39] proposed a multi-objective online matching approach that primarily calculated the Euclidean distance between the current state and predefined objectives. They adjusted the weights of each objective based on the distance. Xu et al. [40] introduced genetic algorithms and simulated annealing algorithms for multi-objective ride-sharing matching, both of which achieved good results. Javidi et al. [42] use biogeography-based optimization to solve a multi-objective optimization problem for online ride-sharing. These methods still face issues of getting trapped in local optima and slow convergence.

Among the mentioned classical dynamic ride-sharing multi-objective optimization methods, only the approach proposed by Xu et al. [40] and Javidi et al. [42] involve evolutionary computation techniques. Hence, there is further room for improvement by incorporating evolutionary computation methods into existing dynamic ride-sharing multi-objective optimization approaches. Table I presents the mentioned classical dynamic ride-sharing multi-objective optimization methods and their optimization objectives.

From the table, it can be observed that the current researches on dynamic ride-sharing optimization primarily focus on reducing detour costs and increasing response rate, with less emphasis on improving user similarity. Furthermore, there is limited involvement of evolutionary computation methods in existing studies. In this research, we need to analyze social relationship network using graph neural network techniques, calculate user similarity as an optimization objective, and obtain the optima through evolutionary computation.

B. Social Network Node Classification

Social network analysis has always been popular among scholars, especially in the field of public safety, where it serves as an important intelligence analysis tool for law enforcement agencies [10]. Graph-based analysis methods have overcome limitations of traditional algorithms in addressing

deep-level problems such as node classification and relationship prediction.

Tang et al. [20] put forward a novel method that iteratively analyzes the neighborhood of nodes and infers the category of the central node by calculating a weighted probability label based on the neighboring nodes. Although this method has been proven effective, its computational complexity is too high due to the high sparsity of the adjacency matrix, making it unsuitable for handling large-scale complex network data. To address this issue, Zhang et al. [21] directly used techniques like DeepWalk to generate feature vectors, providing a statistical-based method for complex network node classification prediction using regression models. While this method simplifies computation, its accuracy improvement compared to using DeepWalk alone is not significant. Kipf and Welling [22] successfully applied graph convolutional models in social network to obtain good data representations and demonstrated excellent performance in semi-supervised social network node classification tasks. Zhan et al. [23] achieved information propagation in social network structural data by utilizing attention neural network models. This method combines link information between nodes with node attributes to generate new node features. Subsequently, a classifier is used to train the correspondence between these fused new features and labels, resulting in classification results. Shang et al. [24] introduced malicious nodes in dynamic social network and studied opinion dynamics models in such scenarios to achieve resilient opinion consensus. However, this method does not consider attention mechanisms, which makes it difficult to aggregate neighbor node features. Yang et al. [35] introduced a graph embedding method that integrates location information, designed for location-based social networks. Lorenz et al. [36] proposed an opinion dynamics model based on relatively unanimous belief, analyzing how beliefs and opinions spread within a social network. Yang et al. [44] combined Expectation-Maximization with GCN to enhance node classification by optimizing topology and learning node representations. Veličković et al. [25] introduced an attention mechanism to handle variable-sized inputs, allowing the model to differentiate the importance of different nodes and edges. Min et al. [37] developed a multi-feature representation learning framework that extracts co-occurrence event features through trajectory resonance, capturing complex social interactions. Lastly, Costa et al. [11] utilized an actor-critic method based on deep reinforcement learning for community detection in large-scale social networks.

Table II illustrates a comparison of several classical social network node classification models and their optimization strategies.

Graph neural network and opinion dynamics share similarities in the field of social network analysis. However, there is currently limited research that utilizes opinion dynamics to optimize graph neural network.

III. RIDE-SHARING FRAMEWORK

The dynamic ride-sharing matching problem is a multi-objective optimization problem, and there are different optimization objectives from different emphases. Common

TABLE II
COMPARISON OF CLASSICAL SOCIAL NETWORK NODE CLASSIFICATION MODELS

Method	Multi-features	Attention Mechanism	Opinion Dynamics
ONI [20]	N	N	N
NES [21]	Y	N	N
GCN [22]	N	N	N
DAC [23]	Y	Y	N
OD [24]	Y	N	Y
LBSN2Vec [24]	Y	N	N
RAOD [36]	N	N	N
EM-GCN [44]	Y	N	N
GAT [25]	Y	Y	N
STGRN [37]	Y	N	N
AC2CD [11]	Y	N	N

optimization objectives include detour cost C , response rate R_{res} , and user similarity rate R_{sim} . Definitions are as follows:

Definition 1 (Detour Cost): Use travel time to represent fuel consumption and other cost on the route. In a certain itinerary, the travel cost of a certain vehicle when it completes the original pick-up task can be represented by C_i , and after adding the route for picking up new passengers, the travel cost can be represented by C_j . Fig. 1 is a schematic diagram of the detour cost. Fig. 1(a) shows the original travel path of passenger i , and Fig. 1(b) shows the impact of inserting new passenger j on the original passenger i 's path offset.

Assuming that passenger i departs at time t_0 , arrives at time t_n if there are no other passengers, and arrives at time t'_n if there are passengers who need to be picked up, then the detour cost brought by adding passengers to the original itinerary is calculated using Equation (1):

$$C_{\Delta(i,j)} = C_j - C_i = t'_n - t_n \quad (1)$$

For the detour cost of C_i and C_j mentioned in Equation (1), use the detour cost of Equation (2) to measure:

$$\xi_{\Delta(i,j)} = \frac{C_{\Delta(i,j)}}{C_i} = \frac{t'_n - t_n}{t_n - t_0} \quad (2)$$

Definition 2 (Effective Allocation of Dynamic Ride Sharing): For each request, dynamic ride sharing needs to meet the following conditions before it can be considered as an effective completion of the request:

- 1) The number of passengers requested for each call is less than or equal to the number of vacancies in the car;
- 2) Passengers get on and off the car within the specified window time;
- 3) Detour cost rate is less than the maximum tolerance value.

Definition 3 (User Similarity): User similarity is used to measure the degree of similarity between two users, and is calculated using cosine similarity. Suppose $f(u)$ means classifying user u , for the classification results $f(u_a)$ and $f(u_b)$ obtained by two users u_a and u_b through the online social network classification model, the user similarity R_{sim}

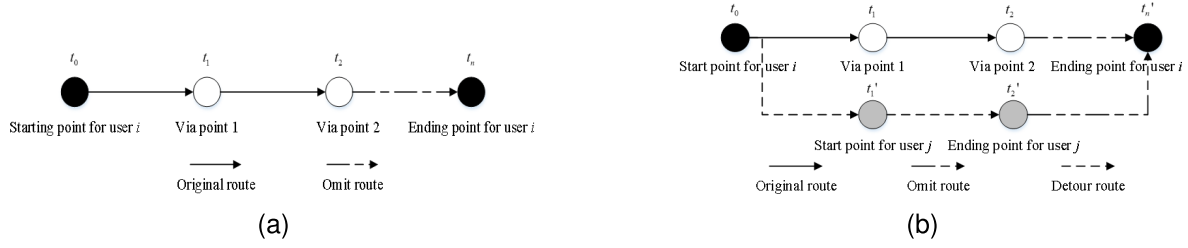


Fig. 1. Schematic diagram of detour cost. (a) The original route of passenger i . (b) Detour cost caused by inserting passenger j .

measurement formula is shown in Equation (3):

$$R_{sim} = \frac{\sqrt{\sum_{i=1}^m (a_i)^2} \sqrt{\sum_{i=1}^m (b_i)^2}}{\sum_{i=1}^m a_i b_i} \quad (3)$$

In the equation, $a_i \in f(u_a) = \{a_1, \dots, a_m\}$ represent the vector of the predicted classification result of u_a , $b_i \in f(u_b) = \{b_1, \dots, b_m\}$ represent the vector of the predicted classification result of u_b , and m is the dimension of the classification result vector.

Definition 4 (Optimization Goal): For a vehicle traveling from v_i to v_j , the total number of car-calling requests in a certain time window $[time_i, time_j]$ is $n_{tr_{total}}$, and the success matching request is $n_{tr_{suc}}$, the optimization goal of the dynamic ride-sharing problem is $Z = \{f_1, f_2, f_3\}$, which can be defined as: minimizing the total detour costs of all vehicles $\sum \xi_\Delta$ within the time window, maximizing the response rate R_{res} , and maximizing the sum of global passenger similarity $\sum R_{sim}$ (evaluated using Equation (3) to examine the similarity between any two passengers within the same vehicle).

$$\begin{cases} \min f_1 = \sum_{i=1}^c \xi_\Delta \\ \max f_2 = R_{res} = \frac{n_{tr_{suc}}}{n_{tr_{total}}} \\ \max f_3 = \sum_{i=1}^m R_{sim} \end{cases} \quad (4)$$

In the equation, it is assumed that there are m passengers and c available vehicles in the entire environment. Some passengers are already in the vehicle, while others are making requests by the roadside, their sum is m . As long as the vehicle is available with reference to definition 2, it can pick up passengers. f_1 ensures that the additional time consumption for each passenger due to ride-sharing is minimized, f_2 ensures that the matching scheme responds to the needs of as many passengers as possible, and f_3 ensures that the pairwise similarity between passengers within each vehicle is maximized.

The optimization goals in the dynamic ride-sharing problem, encapsulated in $Z = \{f_1, f_2, f_3\}$, introduce a complex interaction among the objectives. Minimizing total detour costs f_1 prioritizes efficiency but may compromise response rates f_2 and passenger similarity f_3 . Maximizing response rates enhances accessibility but may increase detour costs

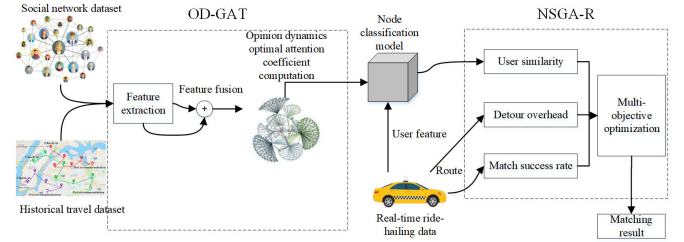


Fig. 2. Overall frame diagram of ride-sharing matching algorithm.

and reduce passenger similarity. Simultaneously, maximizing passenger similarity seeks improved ride quality but may result in longer detours and lower response rates.

IV. GRAPH NEURAL NETWORK AND EVOLUTIONARY COMPUTATION FOR INTELLIGENT RIDE-SHARING RECOMMENDATION

Ride-sharing brings together diverse passengers in a compact and enclosed vehicle, which can present certain safety considerations. Improving service quality and user experience involves matching users with similar attributes, either through clustering or classification. This assumption serves as the basis for our work. Building on this, we propose that grouping users classified as “solo female travelers” or “minor passengers,” for instance, can enhance safety [19], [43]. However, users in online social network are often unwilling to disclose too much private information, and user feature extraction is very difficult. Therefore, it is necessary to study feature extraction and classification through the network topology.

However, it is uneconomical and unrealistic to only consider the degree of similarity between users [28], [29]. If only the similarity between users is considered, the ride-sharing problem is in the worst case, and degenerates into an ordinary taxi route planning problem with one passenger.

The dynamic ride-sharing algorithm that takes safety and efficiency into account is a multi-objective optimization problem, it needs to select the passengers who are on the way and have the highest similarity with the passengers in the vehicle while the route offset is the smallest, and ensure the waiting time of most passengers globally is all within the threshold. The overall framework of the ride-sharing matching algorithm is shown in Fig. 2.

A. Graph Attention Network With Opinion Dynamics

In view of the fact that the existing researches are limited to the analysis and mining of a single relationship network, there

is a lack of multi-dimensional information mutual assistance, verification and fusion analysis methods, and the existing attention mechanism is difficult to learn and simulate the spread of opinions and popularity in the community [26], [27]. To estimate the magnitude of the mutual influence of individuals in the decision-making process, this study proposes a social analysis model of Graph Attention Network with Opinion Dynamics (OD-GAT). It adopts opinion dynamics to optimize the attention mechanism, which is more sensitive, can simulate the size of an individual's influence by others in decision-making, and is more suitable for complex social relationship network; it fully integrates multi-dimensional information such as statistics and embedding for reasoning, making users divided based on factors such as age group and gender according to the predicted labels. OD-GAT can be divided into the following 5 layers:

- 1) Raw data input layer: input the adjacency matrix composed of social network and trajectory network.
- 2) Embedding fusion layer: convert the feature vector and adjacency relationship of nodes into a low-dimensional space vector representation, and perform multi-feature fusion at the same time to input subsequent layers.
- 3) Graph attention convolution layer: A layer that utilizes the attention mechanism to model and weight the relationship between nodes to spread and aggregate information with more targeted process.
- 4) Attention update layer: The role of the attention update layer is to calculate new attention coefficients through an iterative update process based on the state of the current node and the opinions of neighbor nodes.
- 5) Output layer: Mapping the node representations learned by the graph attention network to node classification.

In OD-GAT, firstly, we need to extract the embedded features and statistical features of user social network and trajectory network [30], [31]. Embedded features are extracted using the Node2Vec algorithm [32]. Node statistical features include degree features, that is, for each node, calculate its in-degree and out-degree, and use it as one of the node's feature vectors; neighbor node features, that is, by aggregating the features of the node's neighbors, obtaining the statistical vector of the neighbors' features; subgraph features, that is, divide the graph into multiple subgraphs, calculate the feature vector of the sub-graph it is in, and use it as one of the feature vectors of the node. The trajectory network utilizes Min's proposed method of trajectory resonance for mining, similar to frequent pattern mining. It assigns higher weights to events that co-occur at greater intervals, akin to discovering hidden relationships between users [30]. Since the feature dimension extracted from the network datasets are usually not high, OD-GAT considers early weighted fusion to make the calculation more streamlined [33].

To enable the node's attention coefficient to be dynamically updated to reflect the interaction between nodes, OD-GAT uses the Hegselmann-Krause model to optimize the calculation of the attention coefficient. One of the characteristics of this model is that during the process, the opinion values of each dimension of the nodes will gradually tend to approach each

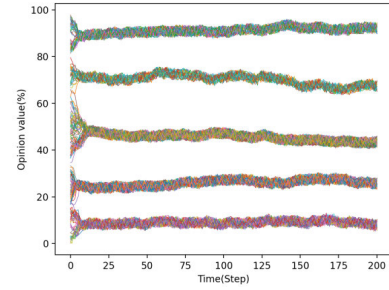


Fig. 3. The evolution of opinion value on Gowalla.

other, and finally converge to a common opinion value. This is because at each time step, each node communicates with randomly selected neighbor nodes and updates its own opinion value. During this update process, the opinion value of the node will be affected by the neighbor nodes and move closer to the opinion value of the neighbor nodes. At the same time, each node will also add a small random disturbance to the opinion value to maintain a certain diversity. After a long enough time, the opinion values of the nodes will gradually converge to the same value or close to the value [41].

Fig. 3 shows how the opinion value of a certain dimension in the opinion vector changes with time steps on a real social network generated by the Gowalla [34]. The opinion value converges to multiple values. This is because the topological structure of the network in the real social network is complex and diverse, and nodes can only be affected by nodes within a certain range of hops or within a certain opinion difference. Therefore, in order to meet the diversity of opinions to be realistic, OD-GAT uses the Hegselmann-Krause model to update the attention coefficient, which only needs to consider a limited time step.

For a given graph G , the attention coefficient of node i for j has been calculated as $\alpha_{i,j}$, and we use the attention coefficient of node i as its initial opinion vector $o_i(t)$, shows in Equation 5, expressing its opinion on a certain issue:

$$o_i(t) = \{\alpha_{i,j}\} \quad (5)$$

The opinion vector $o_i(t)$ of each node i can be updated by aggregating the opinion vectors of neighbor nodes. The updated opinion vector is $o_i(t+1)$, shows in Equation 6:

$$o_i(t+1) = \frac{\sum_{|o_i(t)-o_j(t)| < \varepsilon} o_j(t)}{|N_i|} \quad (6)$$

Here, N_i represents the set of neighboring nodes of node i , and $|N_i|$ represents the count of neighboring nodes. The equation describes the update rule where $o_i(t+1)$, the updated opinion vector of node i , is computed as the average opinion of its neighbors within a certain opinion difference threshold ε . After k iterations, we can determine the attention coefficients α' optimized through opinion dynamics, shows in Equation 7.

$$\alpha'_{ij} = o_i(k) \quad (7)$$

The OD-GAT framework dynamically updates these attention coefficients, adapting the interaction among nodes to enhance the model's performance. The convergence of opinion

values to a common or proximal value over time, as depicted in Fig. 3, illustrates the dynamics of this model in a real social network scenario generated by Gowalla [34]. For a given graph G , with computed attention coefficients α_{ij} for node i with respect to j , we use the attention coefficient α_{ij} of node i as its initial opinion vector $o_i(t)$. This vector represents its perspective on a certain question, and thus the y -axis corresponds to the values of attention coefficients, which are dimensionless.

By introducing Hegselmann-Krause model, the learning object is still to learn node representations that capture contextual information within the graph structure, enabling accurate node classification. The loss function for OD-GAT employs the cross-entropy loss.

B. Non-Dominated Sorting Genetic Algorithm for Ride-Sharing

In view of the fact that the existing researches have little consideration for the optimization objective of humanity factors of user similarity, and the multi-objective optimization process is trapped in local optima and difficult convergence, this study proposes the Non-dominated Sorting Genetic Algorithm for Ride-sharing (NSGA-R).

When inputting Equation 4 as an optimization target for NSGA-R processing, it needs to be modified to some extent. Assuming that there are m passengers in the situation, we need to calculate the average routing similarity, average response rate, and average user similarity:

The average routing similarity F_1 measures the average of the similarity of paths between matched passengers when the optima are reached. Since the travel distances of different passengers vary greatly, the similarity to the matched passenger path needs to be based on its own original path. The average routing similarity F_1 is negatively correlated with the detour cost f_1 of the optimization target, and the calculation equation is as follows:

$$\max F_1 = \frac{\sum_{i=1}^m (100 - \Delta\xi)}{m} \quad (8)$$

The average response rate F_2 is measured as the average of the proportion of passengers served to the total passengers making requests in multiple time windows. The average response rate F_2 is positively correlated with the response rate f_2 of the optimization target, and the calculation equation is as follows:

$$\max F_2 = \frac{f_2}{m} \quad (9)$$

The average user similarity rate F_3 measures the average similarity between matching passengers when the optima are reached. The average user similarity F_3 is positively correlated with the user similarity f_3 of the optimization target, and the calculation equation is as follows:

$$\max F_3 = \frac{f_3}{m} \quad (10)$$

For the designed objective function, it can be directly input into NSGA-R, and the obtained Pareto optima set is a

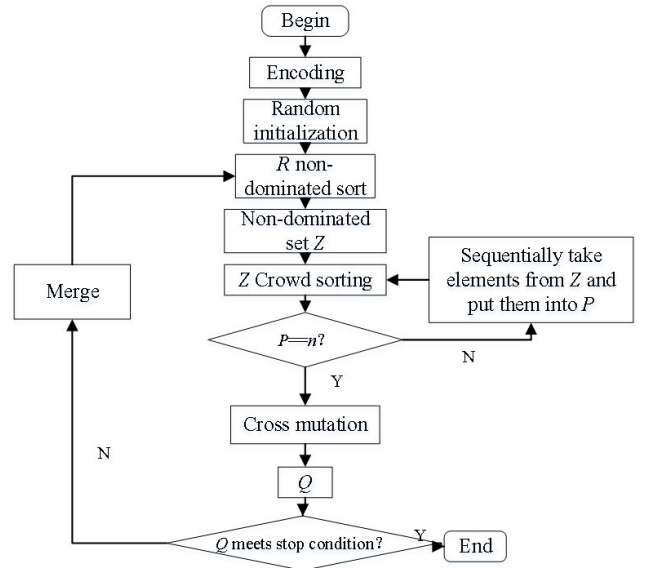


Fig. 4. NSGA-R flow chart.

TABLE III
OVERVIEW OF SNAP

	Gowalla	Brightkite
User (Node)	1.97×10^5	5.82×10^4
Effective User (Trajectory data > 0)	1.07×10^5	5.07×10^4
Relationship (Edge)	9.50×10^5	2.14×10^5
Trajectory Data	6.44×10^6	4.49×10^6
Average User Trajectory Data	60.16	88.61

three-dimensional vectors set [38]. The flowchart of NSGA-R is shown in Fig. 4. Finally, a set of non-dominated optima can be obtained, and the candidate optima represent the optimized ride-sharing scheme, which meets the requirements of minimum detour cost, maximum response rate within the time window, and maximum similarity between passengers.

The multi-objective ride-sharing matching problem was initially an NP-hard issue. However, in NASA-R, its complexity has been reduced to the most time-consuming step, the Crowding Distance Calculation, with a complexity of $O(\text{population size} \times \text{front size})$. By controlling the front size, it can be transformed into a problem solvable in polynomial time.

V. EXPERIMENTAL RESULTS

A. Experimental Setup

The datasets used in the experiment are Gowalla and Brightkite [34] under the Stanford Network Analysis Project (SNAP). The dataset used in the experiment includes nodes, the connections between nodes, and the trajectories of nodes (including timestamps), as shown in Table III.

Originally, it was an unlabeled dataset, but Min [30] added classification labels to it. We treat nodes as passengers and their movement trajectories as ride-hailing or travel data. By partitioning the data based on a specific timestamp, we obtain historical journeys and real-time data. The research mainly carried out the following two experiments:

- 1) The influence of opinion dynamics time step on model performance: to explore whether the graph attention network optimized by opinion dynamics can achieve better performance in terms of indicators as the opinion dynamics time step continues to advance, and whether the indicators are improved compared with the baseline.
- 2) Changes of the optima set of the optimization target with the number of iterations: as the algorithm's number of iterations increases, explore how the optima set of the optimization target of the average routing similarity, average response rate, and average user similarity changes. At the same time, the experiment compares the Pareto optima sets between models, and explores whether the Pareto optima set of this method exhibits superior performance compared to the comparison method based on various indicators.

B. Analysis of Social Network Node Classification

For the social network classification problem, LBSN2Vec [35], RAOD [36], EM-GCN [44], GAT [25], STGRN [37], AC2CD [11] are selected as baseline algorithms. The followings are brief introduction to these comparative algorithms:

LBSN2Vec [35] A graph embedding method that integrates location information, designed for location-based social network.

RAOD [36] An opinion dynamics model based on relatively unanimous belief.

EM-GCN [44] A method combining the Expectation-Maximization with GCN distinguishes nodes from different communities, while optimizing topology and learning node representations for classification.

GAT [25] The attention mechanism can handle variable-sized inputs, simulate the degree of attention the human brain pays to a certain information.

STGRN [37] A multi-feature representation learning framework for social relationship reasoning, which extracts co-occurrence event features through trajectory resonance.

AC2CD [11] An actor-critic method based on deep reinforcement learning for community detection in large-scale social networks.

OD-GAT (Ours) A graph attention network model based on opinion dynamics, using opinion dynamics to optimize the attention mechanism.

Select Precision (P), Recall (R) and F1-Score ($F1$) as the evaluation indicators.

Firstly, we explore whether OD-GAT can achieve better performance in terms of indicators as the time step of opinion dynamics continues to advance, and we need to find out the time step that improves the model performance the most. When the opinion dynamics time step is 0, it degenerates into the GAT algorithm proposed by Veličković [25].

In order to exclude the impact of special datasets, the same set of experiments was performed on both the Gowalla and Brightkite datasets. In the experiment, for each time step, multiple experiments are performed, and the results are averaged. Fig. 5. shows the experiments on Gowalla and Brightkite. The abscissa is the number of time steps, and the ordinate is the corresponding index.

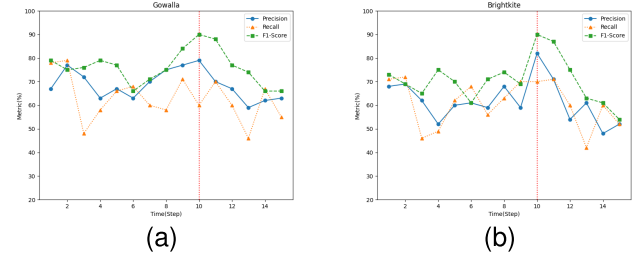


Fig. 5. Experimental results of the influence of opinion dynamics time step on model performance. (a) Experiments performed on the Gowalla. (b) Experiments conducted on the Brightkite.

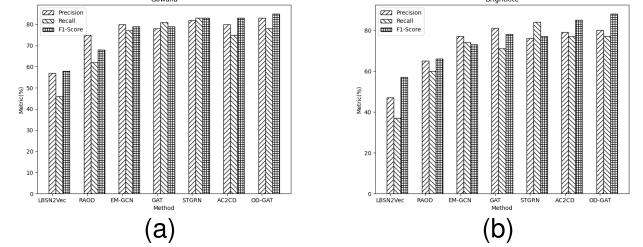


Fig. 6. Experimental results of social node classification performance comparison of different models. (a) Experiments performed on the Gowalla. (b) Experiments conducted on the Brightkite.

The experimental results show that, the performance of the model is improved when the time step is a certain value. This is because in the first few time steps, the model is learning the relatedness and information propagation process between nodes. However, when the time step is too large, too much noise or information duplication may be introduced, resulting in model performance degradation. When the time step is a certain value, the capacity of the model matches the size and complexity of the dataset, so that it can better capture the key features of the node classification problem.

Moving forward, we investigate whether OD-GAT can outperform other classical models. In the previous experiment, when the opinion dynamics evolution time step was set to 10, OD-GAT achieved the best performance. Therefore, in this experiment, the opinion dynamics evolution time step was set to 10; to exclude the impact of special datasets, we still use Gowalla and Brightkite, and performs the same experiments on them respectively. Compared with the classic models LBSN2Vec [35], RAOD [36], EM-GCN [44], GAT [25], STGRN [37], AC2CD [11] for social network node classification, the results are shown in Fig. 6.

C. Analysis of Multi-Objective Optimization

Regarding the multi-objective optimization problem of ride-sharing matching, MA [39], BBO [42] and SAGA [40] are classic multi-objective optimization algorithms. Here is a brief introduction to these algorithms:

MA [39] An adaptive matching strategy that enables multi-objective optimization by adaptively adjusting the weights assigned to each objective.

SAGA [40] A dynamic ride-sharing matching algorithm based on genetic algorithm and simulated annealing algorithm.

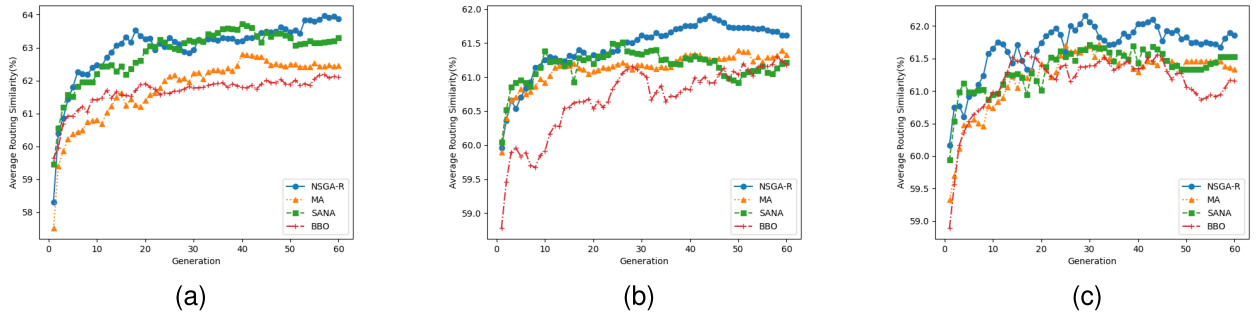


Fig. 7. The optimization process of average routing similarity. (a) 10 passengers. (b) 100 passengers. (c) 1000 passengers.

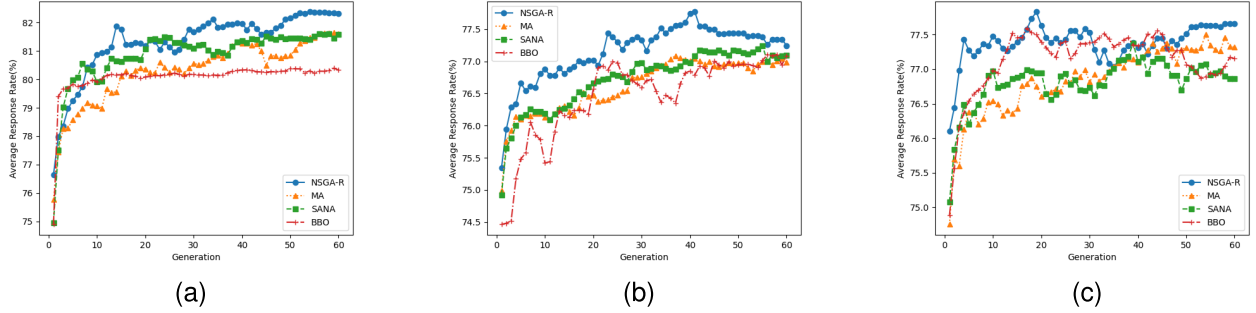


Fig. 8. The optimization process of the average response rate. (a) 10 passengers. (b) 100 passengers. (c) 1000 passengers.

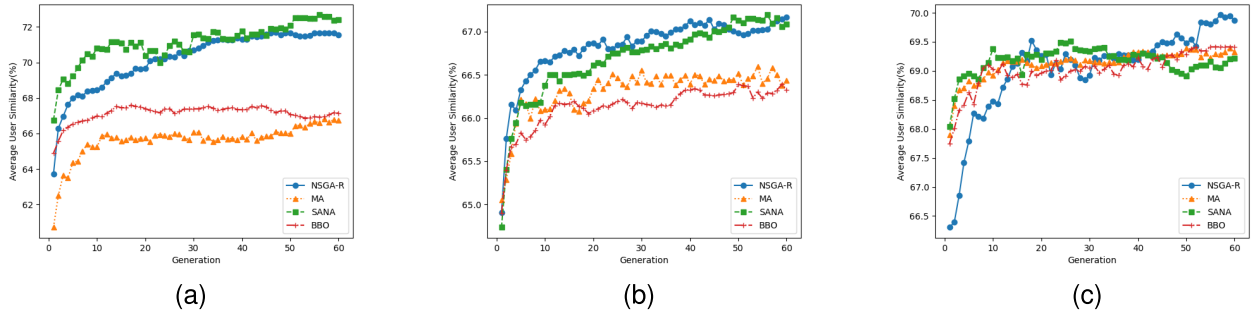


Fig. 9. The optimization process of the average user similarity. (a) 10 passengers. (b) 100 passengers. (c) 1000 passengers.

BBO [42] An algorithm using biogeography-based optimization to solve a multi-objective optimization problem for online ride-sharing.

NSGA-R (Ours) A multi-objective optimization algorithm based on evolutionary computation, searches for the optima by simulating the process of natural selection and genetic operation. This research applies it to multi-objective optimization for ride-sharing matching for the first time.

The algorithms' performance is evaluated based on their convergence speed and convergence value of average routing similarity, average response rate and average user similarity. During the optimization process, we set the initial population size to 100 and the maximum number of iterations to 60. The probability of hybridization is set to 1.0%, and the probability of mutation is set to 0.4%. The number of vehicles in the scene is adjusted to be 70% of the total number of passengers. Throughout the optimization process, changes in average routing similarity, average response rate, and average user similarity were systematically observed and documented. These changes are illustrated in Fig. 7 to Fig. 9.

Fig. 7 shows how the average routing similarity varies with the number of iterations. Fig. 7(a) shows the case of a small sample where the number of passengers is 10. The average routing similarity of the three algorithms increases rapidly from the lower value of the first-generation random population, but the evolutionary computation-based methods NSGA-R and SANA grow faster, converge earlier, and have a higher convergence value than MA; this shows that NSGA-R and SANA have similar performance and are better than MA and BBO in the face of the average routing similarity index of small sample data. Fig. 7(b) and 7(c) show the case of a large sample where the number of passengers is 100 and 1000. The growth rate of the average routing similarity of the three algorithms is similar, and it grows slower than that of the small sample. SANA has a situation where it has not converged after 50 iterations, and the MA and BBO convergence value is relatively low. Although the convergence time of NSGA-R is later than that of small samples, the convergence value is still the highest; this shows that SANA does not perform well on the average routing similarity index for large sample data, the

generalization ability is poor, and the quality of MA optima is not good. NSGA-R has good generalization ability and optima quality.

Fig. 8 shows how the average response rate varies with the number of iterations. Fig. 8(a) shows the case of a small sample of 10 passengers. The average response rates of the three algorithms all increase rapidly from the lower value of the first-generation random population and converge to a similar value; this shows that in the face of the average response rate index of small sample data, the three algorithms have relatively similar performance. Fig. 8(b) and 8(c) show the case of a large sample of 100 and 1000 passengers. The average response rate of NSGA-R shows a slightly higher growth rate compared to SANA and MA. Additionally, the convergence value of NSGA-R is also higher than that of the comparison algorithm. These results indicate that when considering the average response rate index, NSGA-R demonstrates better generalization ability and overall performance compared to SANA, BBO, and MA.

Fig. 9 shows how the average user similarity varies with the number of iterations. Fig. 9(a) shows the small sample situation where the number of passengers is 10. MA falls into local optima prematurely. The performance of SANA in small sample data is similar to that of NSGA-R, both of which are much higher than MA; this shows that in the face of small sample data, the average user similarity index of the data, SANA and NSGA-R have better global search capabilities and avoid falling into local optima. Fig. 9(b) and 9(c) show the case of a large sample of 100 and 1000 passengers. MA still falls into local optima prematurely, and the convergence speed and convergence value of NSGA-R are higher than those of SANA, BBO, and MA; this shows that in the face of the average user similarity index of large samples, only NSGA-R has better generalization ability and global search ability.

In the ride-sharing matching scenario, the number of passengers who concurrently make ride-sharing requests per unit time is generally between 10 and 1000. According to the experimental analysis, compared with MA, BBO and SANA, the NSGA-R algorithm has a stronger global search ability, can cover a wider solution space, can avoid falling into local optima, and requires the least time to generate frontier optima; NSGA-R can still maintain good performance facing a small number of concurrent requests and high-peak concurrent requests in ride-sharing scenarios, and has a certain generalization ability.

VI. CONCLUSION

This research proposes a solution to the safety hazards and multi-objective optimization problems in ride-sharing services, and proves the relevant performance through experiments. Firstly, by introducing opinion dynamics, the Graph Attention Network with Opinion Dynamics (OD-GAT) is proposed, which can more accurately reason about social relations, and improve service quality and safety for ride-sharing users. Secondly, a novel evolutionary computation method for ride-sharing is proposed, which achieves a balance among multiple objectives such as average routing similarity, average response rate, and average user similarity in the

ride-sharing process, and fills the gap between evolutionary calculations in ride-sharing matching optimization.

In conclusion, this research provides useful exploration and solutions for the development of the ride-sharing industry. Additional research can delve into the development of methods to safeguard user privacy while maintaining high service quality, efficiency and personalization of ride-sharing services and provide users with a better travel experience.

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